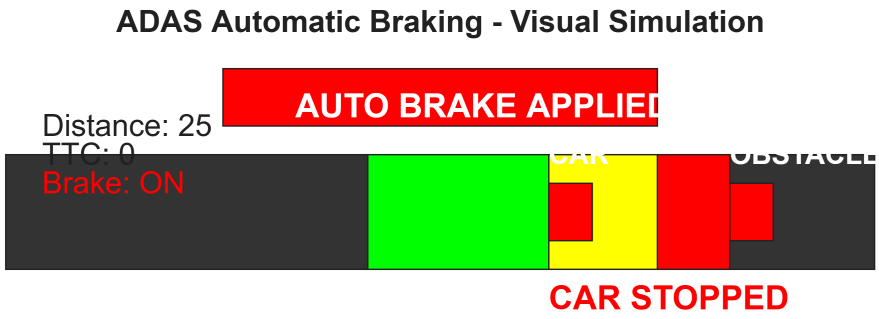
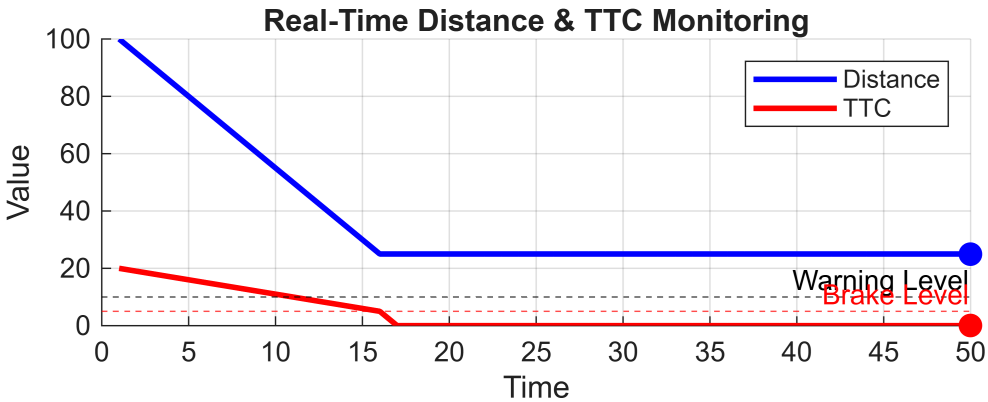
```
clc;
clear;
close all;

car_pos = 0;
obs_pos = 100;
speed = 5;

time = 1:50;

distance_array = zeros(1,50);
ttc_array = zeros(1,50);

figure;

for t = 1:50

    % Distance & TTC
    distance = obs_pos - car_pos;

    if speed ~= 0
        TTC = distance / speed;
    else
        TTC = 0;
    end

    distance_array(t) = distance;
    ttc_array(t) = TTC;

    % Default
    car_color = 'g';
    status = 'SAFE';
    brake_status = 'Brake: OFF';

    % ADAS Logic
    if TTC < 10 && TTC > 5
        car_color = 'y';
        status = 'WARNING ZONE';

    elseif TTC <= 5
        car_color = 'r';
        status = 'DANGER - AUTO BRAKE';
        brake_status = 'Brake: ON';
        speed = 0;
    end

    % Update position
    car_pos = car_pos + speed;

    clf;
```

```
    % ===================== TOP VIEW =====================
    subplot(2,1,1);
    hold on;

    % Road
    rectangle('Position',[0 -2 120 4],'FaceColor',[0.2 0.2 0.2]);

    % Zones
    rectangle('Position',[obs_pos-50 -2 50 4],'FaceColor',[0 1 0 0.1]); %
Safe
    rectangle('Position',[obs_pos-25 -2 25 4],'FaceColor',[1 1 0 0.2]); %
Warning
    rectangle('Position',[obs_pos-10 -2 10 4],'FaceColor',[1 0 0 0.3]); %
Danger

    % Car
    rectangle('Position',[car_pos -1 6 2],'FaceColor',car_color);
    text(car_pos, 2, 'CAR','Color','w','FontWeight','bold');

    % Obstacle
    rectangle('Position',[obs_pos -1 6 2],'FaceColor','r');
    text(obs_pos, 2, 'OBSTACLE','Color','w','FontWeight','bold');

    % Info Display
    text(5, 3, ['Distance: ', num2str(distance)], 'FontSize', 11);
    text(5, 2, ['TTC: ', num2str(TTC)], 'FontSize', 11);
    text(5, 1, brake_status, 'FontSize', 11, 'Color','r');

    % STATUS BOX (VERY CLEAR)
    if TTC <= 5
        rectangle('Position',[30 3 60 2],'FaceColor','r');
        text(40, 3.7, 'AUTO BRAKE APPLIED!',
'Color','w','FontSize',12,'FontWeight','bold');
    elseif TTC < 10
        rectangle('Position',[30 3 60 2],'FaceColor','y');
        text(40, 3.7, 'WARNING: SLOW DOWN',
'Color','k','FontSize',12,'FontWeight','bold');
    else
        rectangle('Position',[30 3 60 2],'FaceColor','g');
        text(40, 3.7, 'SAFE DRIVING',
'Color','w','FontSize',12,'FontWeight','bold');
    end

    % Car stopped indicator
    if speed == 0
        text(car_pos, -3, 'CAR STOPPED',
'Color','r','FontSize',12,'FontWeight','bold');
    end
```

```
    title('ADAS Automatic Braking - Visual Simulation');
    xlim([0 120]);
    ylim([-4 6]);
    axis off;

    % ===================== GRAPH =====================
    subplot(2,1,2);
    hold on;

    plot(time(1:t), distance_array(1:t), 'b', 'LineWidth', 2);
    plot(time(1:t), ttc_array(1:t), 'r', 'LineWidth', 2);

    % Mark current point
    plot(t, distance, 'bo', 'MarkerSize', 8, 'MarkerFaceColor','b');
    plot(t, TTC, 'ro', 'MarkerSize', 8, 'MarkerFaceColor','r');

    % Threshold lines
    yline(10, '--k', 'Warning Level');
    yline(5, '--r', 'Brake Level');

    legend('Distance','TTC','Location','northeast');
    xlabel('Time');
    ylabel('Value');
    title('Real-Time Distance & TTC Monitoring');
    grid on;

    pause(0.3);
end
```


ADAS Automatic Braking - Visual Simulation
AUTO BRAKE APPLIED
Distance: 25
TTC: 0
Brake: ON
CAR STOPPED


Real-Time Distance & TTC Monitoring
Value
Time
Distance
TTC
Warning Level
Brake Level
0
20
40
60
80
100
0
5
10
15
20
25
30
35
40
45
50